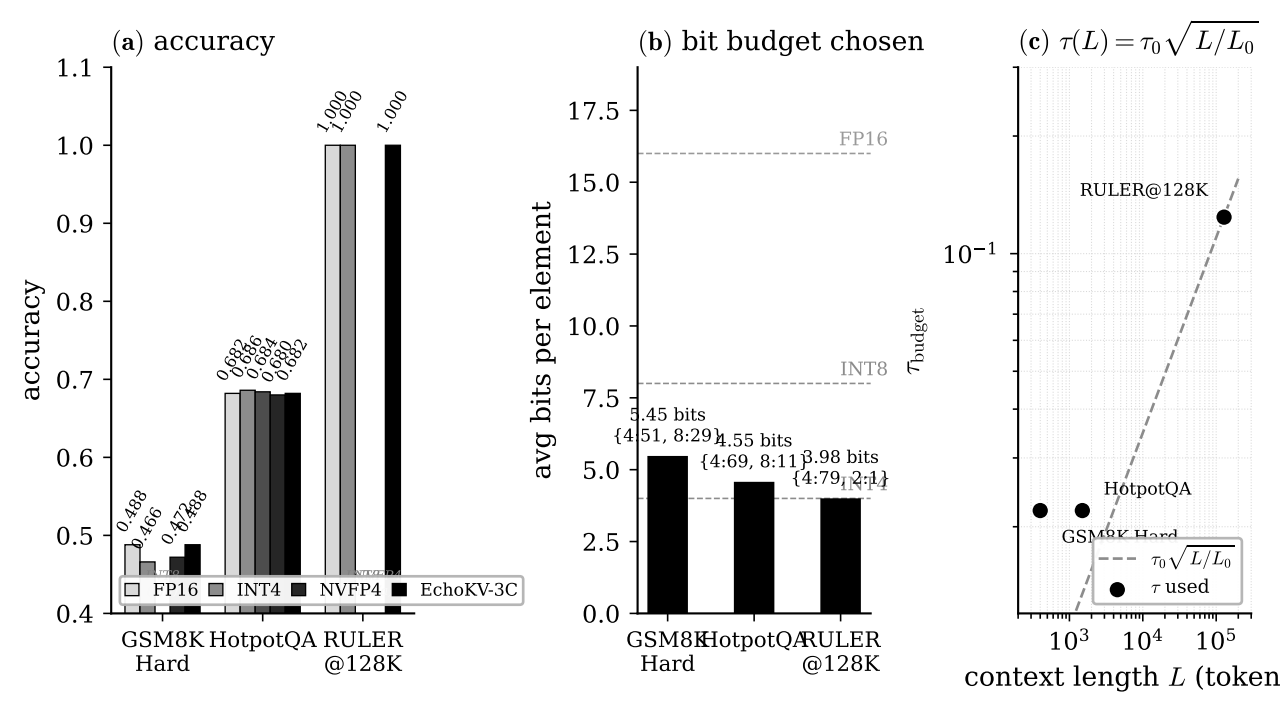
(a) accuracy
accuracy
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.488
0.466
0.472
0.488
0.682
0.686
0.684
0.680
0.682
1.000
1.000
1.000
FP16
INT4
NVFP4
EchoKV-3C
GSM8K Hard
HotpotQA
RULER @128K
(b) bit budget chosen
avg bits per element
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
FP16
INT8
INT4
5.45 bits
{4:51, 8:29}
4.55 bits
{4:69, 8:11}
3.98 bits
{4:79, 2:1}
GSM8K Hard
HotpotQA
RULER @128K
(c) $\tau(L) = \tau_0 \sqrt{L/L_0}$
τ_{budget}
10^{-1}
RULER@128K
HotpotQA
GSM8K Hard
$\tau_0 \sqrt{L/L_0}$
τ used
10^3
10^4
10^5
context length L (token